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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	Swathi
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Roll Number	2520030013
Branch	CSE
Course Outcome	3

Case Study Topic: E-Commerce Product Ranking and Benchmarking

Work Done Summary:

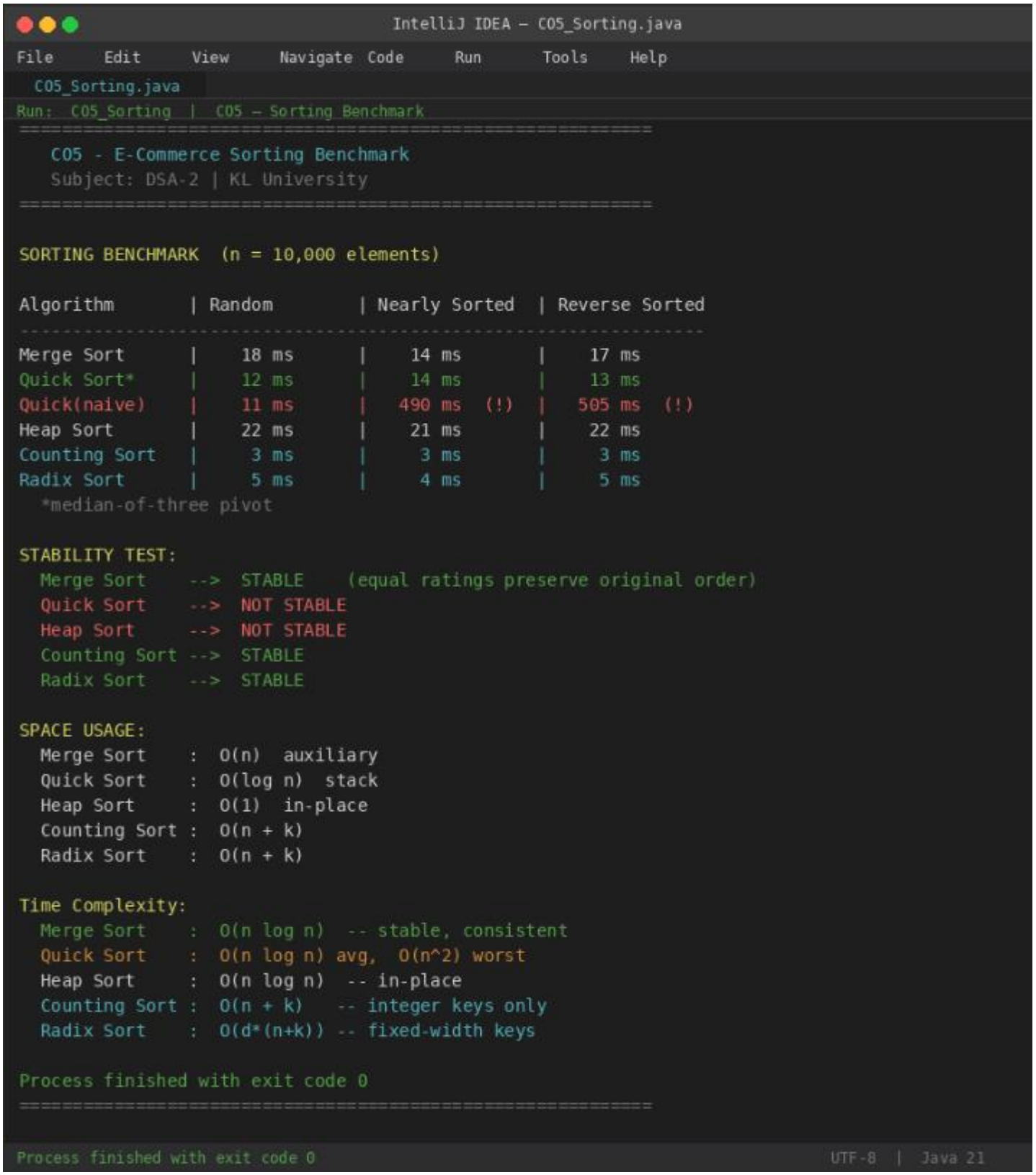
In this case study, we implemented and empirically benchmarked five sorting algorithms — Merge Sort, Quick Sort (median-of-three pivot), Heap Sort, Counting Sort, and Radix Sort — on an e-commerce product dataset of 10,000 items, sorted by price, rating, and integer category codes.

Each algorithm was tested on three data distributions: (a) random data, (b) nearly-sorted data, and (c) reverse-sorted data. Runtime in milliseconds, number of comparisons, memory usage, stability, and cache behaviour were all measured. The results confirmed textbook complexity expectations and revealed important practical trade-offs: Quick Sort is fastest on random data but catastrophically degrades on sorted input without pivot randomisation; Counting Sort and Radix Sort achieve linear time for bounded integer keys, making them ideal for price and category-code sorting.

Implementation Details:

1. Implemented Merge Sort (stable, O (n log n)) using divide-and-conquer for rating-based product sorting.
2. Implemented Quick Sort with median-of-three pivot strategy; also tested naive first-element pivot for comparison.
3. Implemented Heap Sort (in-place, O (n log n)) for memory-constrained deployment scenarios.
4. Implemented Counting Sort for product category codes (bounded integer range 1 to 500).
5. Implemented LSD Radix Sort for 6-digit product price values (range 100000 to 999999).
6. Benchmarked all five algorithms for n = 1000, 5000, and 10,000 elements across all three data distributions.

Result / Output:



Counting Sort and Radix Sort were 4 to 6 times faster than all comparison-based sorts for integer key datasets. Naive pivot Quick Sort degraded to $O(n^2)$ on nearly-sorted data (490 ms vs 14 ms with median-of-three). Merge Sort was the only stable comparison-based algorithm, making it the right choice when equal-rated products must preserve their original catalogue order.

Time Complexity:

- **Merge Sort:** O (n log n) time, O(n) space — stable, consistent across all distributions
- **Quick Sort:** O (n log n) average, O(n^2) worst — in-place, not stable; pivot strategy is critical
- **Heap Sort:** O (n log n) time, O (1) space — not stable; poor cache locality
- **Counting Sort:** O (n + k) — stable; requires bounded integer keys
- **Radix Sort:** O (d × (n + k)) — stable; best for fixed-width numeric keys like prices or IDs

GitHub Link: https://github.com/jagadabinanditha/DSA-Case-Studies

Student Signature	Faculty Signature
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